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Social Media Sentiment and Early Performance of K-Pop Comebacks

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Final Year Project
Interim Progress Report

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1. Project Aim

This project investigates whether a measurable relationship exists between public sentiment expressed on Twitter and Reddit during K-pop group comebacks and the early performance of those releases on YouTube. Performance is measured through engagement metrics including views, likes, and comments within a defined period following release. Six K-pop groups are used as case studies: aespa, IVE, TWICE, NCT DREAM, SEVENTEEN, and Stray Kids. These groups represent different levels of popularity and international reach.

The aim is to explore whether aggregated sentiment from social media discussion can serve as a useful indicator of early digital music performance, contributing to research in social media analytics and natural language processing.

2. Statement of What You Will Do

This project will develop a sentiment analysis pipeline that collects, processes, and analyses social media data related to the comebacks of six K-pop groups. Data will be gathered from Twitter and Reddit, focusing on English-language posts and comments published within a 14-day period surrounding each group's comeback release date.

The collected text data will be preprocessed using the NLTK Python library, including tokenisation, stopword removal, and stemming. After preprocessing, the text will be converted into numerical features using TF-IDF through the scikit-learn library.

A subset of the dataset will then be manually labelled as positive, negative, or neutral. The labelled data will be exported in ARFF format and imported into WEKA, where two classification algorithms, Naïve Bayes and Support Vector Machine, will be trained and evaluated using 10-fold cross-validation.

The best performing classifier will then be applied to the full dataset to generate sentiment scores for each comeback. These scores will be compared with YouTube engagement metrics including views, likes, and comments collected at 24 hours, 72 hours, and one week after release. Pearson and Spearman correlation coefficients will be used to examine whether a relationship exists between sentiment and performance.

The findings will be presented through an interactive Tableau dashboard visualising sentiment distributions, correlation patterns, and comparisons between groups.

3. Objectives

3.1. Core Objectives

C1. Collect at least 500 English-language posts and comments per comeback from Twitter and Reddit using the Tweepy and PRAW Python libraries within a 14-day window surrounding each group's official release date.

C2. Preprocess all collected data using the NLTK Python library, including tokenisation, stopword removal, and stemming, and convert the text into numerical features using TF-IDF through scikit-learn.

C3. Manually label a minimum of 200 samples per comeback as positive, negative, or neutral and export the dataset in ARFF format for use in WEKA.

C4. Train and evaluate Naïve Bayes and Support Vector Machine classifiers in WEKA using 10-fold cross-validation.

C5. Evaluate classifier performance using accuracy, precision, recall, and F1-score to determine the most reliable model.

C6. Apply the best performing model to generate sentiment scores for all six comebacks.

C7. Collect YouTube engagement metrics and calculate Pearson and Spearman correlations.

C8. Present findings in a Tableau dashboard.

3.2. Advanced Objectives

A1. Expand the dataset to include Reddit posts and comments from group-specific subreddits to enable comparison between sentiment expressed on Twitter and Reddit.

A2. Analyse sentiment trends across the full 14-day window to determine whether sentiment changes over time following a comeback release.

A3. Compare sentiment patterns between girl groups and boy groups to explore whether fanbase demographics influence sentiment expression.

4. Research Summary and Bibliography

4.1. Introduction

The literature reviewed for this project focuses on the use of sentiment analysis and social media data to examine public opinion and digital engagement. These studies provide insight into how online discussion can be analysed using natural language processing techniques to identify behavioural patterns. They also inform the methodological decisions made in this project, particularly the use of machine learning classifiers and the comparison of different sentiment analysis approaches.

4.2. Paper Summaries

Paper 1 – Using Twitter to Predict Chart Position for Songs (Tsiara and Tjortjis, 2020)

Tsiara and Tjortjis (2020) investigated whether activity on Twitter could be used to predict the chart performance of songs. The study analysed correlations between mentions of songs and the artists on Twitter and the final ranking of songs on the Billboard Hot 100 chart. Chart data was collected from the Billboard website, while relevant tweets were gathered using the Twitter Search API. A regression analysis model was used to compare predicted and actual chart positions. The study found a moderate correlation between song mentions and chart performance, while artist mentions showed insignificant results. These findings suggest that social media activity surrounding specific songs can influence their performance. This is relevant to the present project, which investigates the relationship between social media sentiment and the early YouTube performance of K-pop comebacks.

Paper 2 – Comparative Analysis of Naïve Bayes Classifier and Support Vector Machine in Sentiment Analysis of Social Media Data (Aprinastya *et al.*, 2024)

Aprinastya et al. (2024) compared the performance of Naïve Bayes and Support Vector Machine classifiers in analysing multilingual user reviews of the game Genshin Impact. The dataset was preprocessed using tokenisation, stemming, and TF-IDF feature extraction before classification. The study found that Naïve Bayes achieved an accuracy score of 71%, while Support Vector Machine achieved 63%. The authors concluded that Naïve Bayes performed better for this multilingual dataset. This research supports the decision in the present project to compare multiple classification algorithms when analysing sentiment in social media posts.

Paper 3 – Comparison of Naïve Bayes Classifier and Support Vector Machine in Classifying Public Responses on Twitter (Millennianita, Athiyah and Muhammad, 2024)

Millennianita, Athiyah and Muhammad (2024) analysed Twitter sentiment regarding the Kim Garam bullying controversy. Tweets were collected using keywords related to the case through a Python crawler in Google Colaboratory using the Tweepy and Pandas libraries. After preprocessing and TF-IDF feature selection, both Naïve Bayes and Support Vector Machine classifiers were applied. The results showed that Support Vector Machine achieved higher accuracy at 97%, compared with 93% for Naïve Bayes. The contrasting findings between this study and Aprinastya et al. (2024) demonstrate that algorithm performance varies depending on the dataset. This supports the decision to evaluate both classifiers in the present project.

4.3. Bibliography

The following sources were also reviewed and helped inform the broader context of this project. (Reisz, Servedio and Thurner, 2024; Zhang *et al.*, 2025)

Aprinastya, R. *et al.* (2024) 'Comparative Analysis of Naïve Bayes Classifier and Support Vector Machine for Multilingual Sentiment Analysis: Insights from Genshin Impact User Reviews', *JUSIFO (Jurnal Sistem Informasi)*, 10(2), pp. 117–126.

Millennianita, F., Athiyah, U. and Muhammad, A.W. (2024) 'Comparison of naive bayes classifier and support vector machine methods for sentiment classification of responses to bullying cases on Twitter', *Journal of Mechatronics and Artificial Intelligence*, 1(1), pp. 11–26.

Reisz, N., Servedio, V.D.P. and Thurner, S. (2024) 'Quantifying the impact of homophily and influencer networks on song popularity prediction', *Scientific Reports*, 14(1), p. 8929. Available at: https://doi.org/10.1038/s41598-024-58969-w.

Tsiara, E. and Tjortjis, C. (2020) 'Using Twitter to Predict Chart Position for Songs', in I. Maglogiannis, L. Iliadis, and E. Pimenidis (eds) *Artificial Intelligence Applications and Innovations*. Cham: Springer International Publishing, pp. 62–72.

Zhang, J. *et al.* (2025) 'A Study on the Impact of Social Media on K-pop Fans' Behavior and Emotions: A Case Study of Chinese Fans', *Interdisciplinary Humanities and Communication Studies*, 1(3).

5. Reflection on Progress

Work on this project began in February 2026 following an initial supervisory meeting held on 13 February at 10:30 am. During this meeting, my supervisor recommended reviewing peer-reviewed literature to provide a stronger academic foundation for the project. Acting on this recommendation, five sources were identified and reviewed, three of which have been summarised in Section 4 of this report.

Reviewing the literature helped clarify the direction of the project and influenced several methodological decisions. It supported the decision to compare Naïve Bayes and Support Vector Machine classifiers rather than selecting a single method. This comparison will allow the project to determine which algorithm performs better on the collected dataset.

One challenge during this stage of the project has been balancing academic work with personal commitments. Unplanned engagements between 27 February and 1 March resulted in the loss of several working days, which required the project timeline to be adjusted. The literature review stage also took longer than expected, as reading and analysing academic papers proved more time-consuming than initially anticipated.

Despite these challenges, meaningful progress has been made. The Twitter Developer account and application have been created and approved, the project GitHub repository has been established with version control active from the beginning, and the Python development environment has been configured with the required libraries installed. Access to the Reddit API has also been requested.

A follow-up meeting scheduled for 5 March was unfortunately missed due to personal circumstances, and a request to reschedule has since been sent.

Reflecting on this phase of the project, stronger time management at the outset would likely have reduced some of the pressure experienced. Going forward, fixed daily working hours will be maintained to ensure steady progress toward the final submission deadline.

6. Ethical Analysis

6a. Ethical Issues in the Area

Using social media data for sentiment analysis raises several ethical concerns. One major issue is privacy. Even though posts on platforms such as Twitter and Reddit are publicly visible, users typically share content within their own communities and may not expect their words to be collected and analysed in academic research.

Another concern relates to bias in machine learning models. Sentiment classification models learn from training data, which can contain cultural or linguistic biases. This means the model may misinterpret certain groups or styles of communication. In the context of K-pop, this is particularly relevant because fan communities are highly international and often use slang, fandom terminology, or expressive language that general English sentiment models may not interpret correctly.

6b. Ethical Issues Specific to this Project

The main ethical consideration for this project involves using publicly available social media posts without direct consent from individual users. Although the data is accessible to the public, the users who created the posts did not explicitly agree to have their content included in a research study.

There is also an element of subjectivity in the manual annotation stage of the project. Sentiment labels are applied by the researcher, and interpretations can vary. A post that one annotator classifies as positive may be interpreted differently by another. This introduces the possibility that some posts could be labelled in a way that does not fully reflect the original intent of the user.

To reduce potential privacy concerns, usernames and other identifying information will not be stored as part of the dataset used for analysis.

6c. Potential Harms

Several potential harms could arise from this type of research. Incorrect sentiment classification may misrepresent the attitudes of K-pop fan communities, particularly if culturally specific language or sarcasm is misinterpreted by the model.

If the findings were interpreted without context, they could also lead to generalisations about fan behaviour that do not accurately represent the community. There is also a small privacy risk when collecting and storing social media posts. Even when usernames or identifying information are removed, the data still originates from real individuals.

6d. Statement on the Use of Artificial Intelligence

According to the University of Hertfordshire's academic integrity policy, AI tools may be used for proofreading purposes but must not generate original academic content that is submitted as the student's own work.

In line with this policy, AI tools were used only to review grammar and clarity in this report. All analysis, interpretation, and written content were produced independently.

Within the technical component of the project, machine learning algorithms such as Naïve Bayes and Support Vector Machine are used to perform sentiment classification. These algorithms are standard analytical techniques within data science and natural language processing and are part of the research methodology rather than AI-generated writing.

6e. Ethics Approval

This project does not require formal ethics approval from the University of Hertfordshire.

Ethics approval is typically required when research involves direct interaction with participants, collection of sensitive personal data, or research involving vulnerable groups. This project uses only publicly available social media posts and does not involve surveys, interviews, or direct engagement with individuals.

No personally identifiable information is collected or stored as part of the dataset. For this reason, the project falls within the category of research that is exempt from formal ethics review under the university's guidelines.

7. BCS Code of Conduct

Standard 1 – Public Interest (Privacy)

This states that you shall:

a. have due regard for public health, privacy, security and wellbeing of others and the environment.
b. have due regard for the legitimate rights of Third Parties.
c. conduct your professional activities without discrimination on the grounds of sex, sexual orientation, marital status, nationality, colour, race, ethnic origin, religion, age or disability, or of any other condition or requirement.
d. promote equal access to the benefits of IT and seek to promote the inclusion of all sectors in society wherever opportunities arise.

This standard is relevant because the project involves analysing posts written by real social media users. Although the data used is publicly accessible and personal identifiers are not stored, there is still a responsibility to handle the information responsibly. The results must therefore be presented carefully to avoid misrepresenting the individuals or communities whose content is being analysed.

Standard 2 – Professional Competence and Integrity

This states that you shall:

a. only undertake to do work or provide a service that is within your professional competence.
b. NOT claim any level of competence that you do not possess.
c. develop your professional knowledge, skills and competence on a continuing basis, maintaining awareness of technological developments, procedures, and standards that are relevant to your field.
d. ensure that you have the knowledge and understanding of Legislation* and that you comply with such Legislation, in carrying out your professional responsibilities.
e. respect and value alternative viewpoints and, seek, accept and offer honest criticisms of work.
f. avoid injuring others, their property, reputation, or employment by false or malicious or negligent action or inaction.
g. reject and will not make any offer of bribery or unethical inducement.

This standard relates to ensuring that the project is carried out using appropriate methods and tools. The technologies used in this project, including Python for data processing, WEKA for machine learning classification, and Tableau for data visualisation, are widely recognised tools within data science. Using established tools and clearly documented methods helps ensure that the work meets appropriate professional and academic standards.

Standard 3 – Duty to the Profession

This states that you shall:

a. accept your personal duty to uphold the reputation of the profession and not take
b. any action which could bring the profession into disrepute.
c. seek to improve professional standards through participation in their
d. development, use and enforcement.
e. uphold the reputation and good standing of BCS, the Chartered Institute for IT.
f. act with integrity and respect in your professional relationships with all members
g. of BCS and with members of other professions with whom you work in a
h. professional capacity.
i. encourage and support fellow members in their professional development.

This project contributes to research exploring how sentiment analysis can be applied to social media discussion in cultural and entertainment contexts. By following clear research methodology, documenting limitations, and maintaining proper version control throughout the project, the work supports the standards expected within the computing profession and contributes to responsible research practice.

8. Risk Register

The following risk register identifies potential issues that could affect the progress or outcome of the project and outlines steps taken to reduce their impact.

Risk	Likelihood	Impact	Mitigation
Twitter API rate limits restrict the volume of data collected	High	High	Supplement dataset with Reddit posts and prioritise key comebacks
Personal commitments reducing available time for project work	High	High	Maintain fixed working hours and prioritise core tasks
WEKA classification accuracy too low to support meaningful conclusions	Medium	High	Apply 10-fold cross-validation and compare both classifiers

Inconsistent manual data labelling affecting model accuracy	Medium	Medium	Establish clear sentiment labelling guidelines before annotation
Supervisor unavailable close to the deadline	Low	Medium	Maintain detailed documentation and raise questions via email early
Loss of data or code due to technical failure	Low	High	Use regular GitHub commits and maintain both local and cloud backups

9. Project Plan

This project follows an iterative development approach. Each stage builds on the results produced in the previous stage. This structure is suitable for sentiment analysis because the process involves several connected phases, including data collection, preprocessing, model training, and correlation analysis. Each stage may require adjustments based on the results produced in earlier stages of the workflow.

The remaining stages of the project will focus on completing the technical pipeline. The next step involves implementing the data collection scripts using the Twitter and Reddit APIs. Once the dataset has been collected, the data will be preprocessed and manually labelled for sentiment classification. The labelled dataset will then be used to train and evaluate the Naïve Bayes and Support Vector Machine models in WEKA. After classification is complete, the generated sentiment scores will be compared with YouTube engagement metrics using correlation analysis. The final stage of the project will involve presenting the results through an interactive Tableau dashboard.

Version control has been maintained throughout the project using GitHub. Commits have been recorded regularly since the initial project setup in February 2026 to provide a clear history of development progress. The repository structure and commit history are shown in Figure 1.

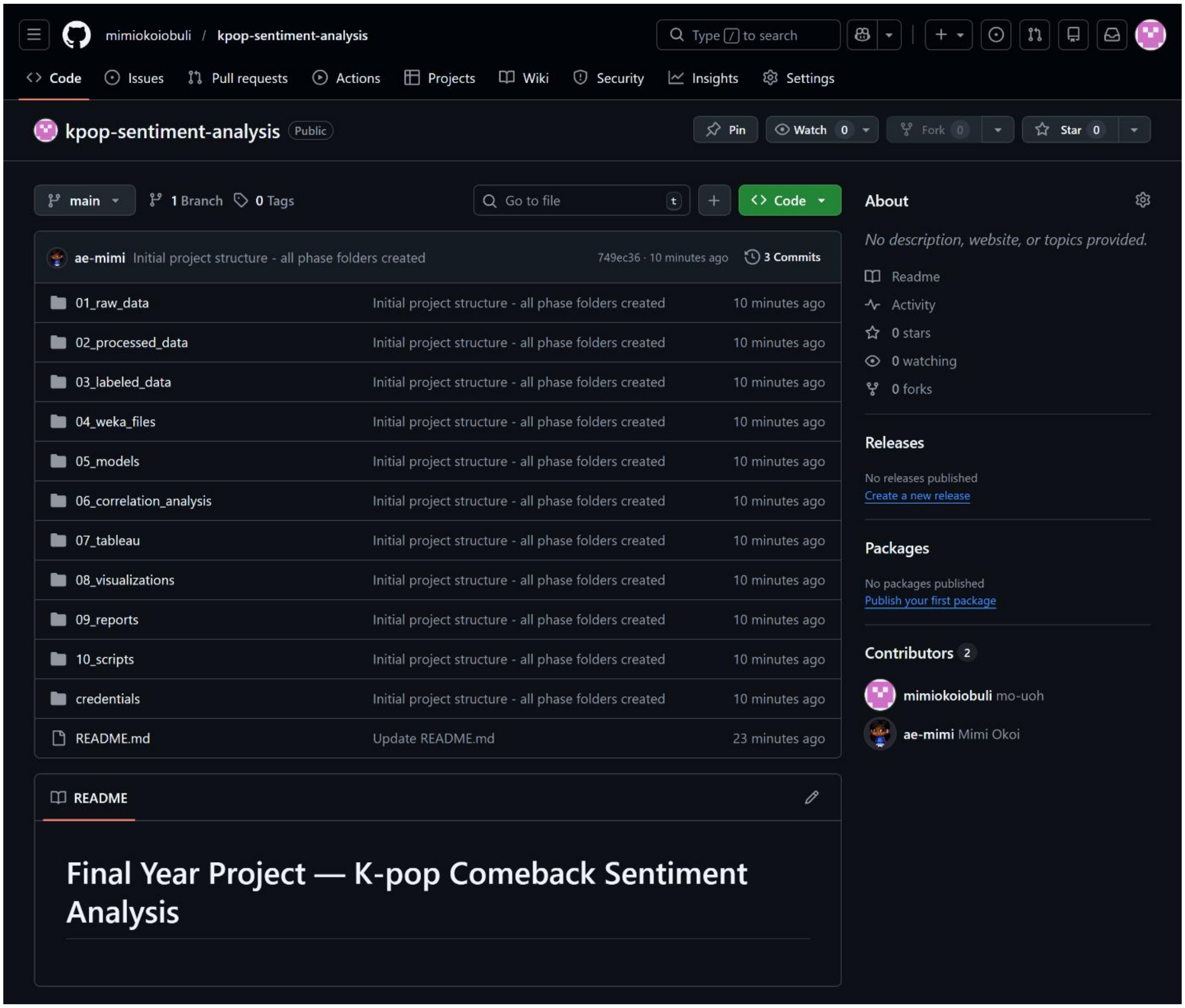

Figure 1. GitHub repository for the project showing the project folder structure and commit history used to track development progress.

10. Way of Working

Harvard referencing has been applied throughout the report in accordance with the *Cite Them Right* 12th edition guidelines. References have been managed using Zotero to ensure consistency and accurate citation formatting. Tasks set for the module, including the literature review and preparation of this interim progress report, have been completed in line with the project schedule.

Version control has been maintained using GitHub, with commits recorded at each major stage of the project. This provides a transparent record of development progress and changes made during the project.

A supervisory meeting took place on 13 February 2026. During this meeting, the recommendation was made to review additional peer-reviewed literature relevant to the research topic. This recommendation was incorporated into the literature review stage of the project and informed the sources summarised in Section 4. A follow-up meeting scheduled for 5 March was unfortunately missed due to personal circumstances, and a request to reschedule has since been sent.

All written content in this report has been produced independently in line with the University of Hertfordshire's academic integrity policy.